FINITE STRATIFICATION OF A QUASI-COMPACT REASONABLE ALGEBRAIC SPACE BY SCHEME PIECES

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Editorial note. Thin delegation card: the source says the proof is identical to the quasi-compact filtration lemma and adds only the termination observation; both the delegation and the observation are carried as explicit obligations.

Allowed standard prerequisites (editorial).

- Quasi-compact filtration construction. lemma-filter-quasi-compact (carded L26712). The construction, run verbatim.
- Reasonable bound. definition-very-reasonable (same chapter, structural). The integer n bounding the fibres of U -> X.

Source reference. Original definition 03I8: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1262–1262. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0BBN: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1750–1750. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact, reasonable algebraic space over S. There exist an integer n and open subspaces*

$$\emptyset = U_{n+1} \subset U_n \subset U_{n-1} \subset \ldots \subset U_1 = X$$

with the following property: setting $T_p = U_p \setminus U_{p+1}$ (with reduced induced subspace structure) there exists a separated scheme V_p and a surjective étale morphism $f_p : V_p \to U_p$ such that $f_p^{-1}(T_p) \to T_p$ is an isomorphism.

Proof. The proof of this lemma is identical to the proof of Lemma 0BBN. Let n be an integer bounding the degrees of the fibres of $U \to X$ which exists as X is reasonable, see Definition 03I8. Then we see that $U_{n+1} = \emptyset$ and the proof is complete. □

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.